

CHISINEU-CRIS, Romania

WMO: 151360

Lat: 46.5186N

Lon: 21.5417E

Elev: 97

StdP: 100.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.4	-12.8	1.3	-9.6	-10.2	1.6	-7.1	10.6	-2.8	9.4	-0.7	2.3	0	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	34.0	20.6	32.4	20.8	30.6	20.3	23.1	28.6	22.2	28.6	21.5	27.9	2.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.6	24.6	20.6	15.5	23.8	19.7	14.6	22.9	68.8	28.2	65.6	28.6	62.8	28.2	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.6	7.6	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-0.8	2.8	6.1	11.7	16.9	20.7	21.7	22.4	17.4	11.7	7.0	1.9	(6)
(7)		DBStd	8.86	4.62	4.67	4.41	4.69	3.43	3.22	2.99	3.36	4.11	4.34	4.45	4.28	(7)
(8)		HDD10.0	1109	336	203	132	38	2	0	0	0	1	34	111	253	(8)
(9)		HDD18.3	2824	594	435	378	202	67	13	6	5	68	207	339	510	(9)
(10)		CDD10.0	1717	0	2	12	87	217	320	364	384	221	87	22	1	(10)
(11)		CDD18.3	392	0	0	0	2	23	84	112	131	38	2	0	0	(11)
(12)		CDH23.3	4432	0	0	1	53	303	889	1217	1496	429	44	0	0	(12)
(13)		CDH26.7	1736	0	0	0	3	62	320	491	708	150	2	0	0	(13)
(14)	Wind	WSAvg	2.6	2.7	3.2	3.2	3.1	2.8	2.3	2.3	2.2	2.1	2.2	2.6	2.8	(14)
(15)	Precipitation	PrecAvg	592	35	30	32	48	57	71	69	47	60	53	47	45	(15)
(16)		PrecMax	847	79	92	145	89	137	189	143	105	141	125	113	103	(16)
(17)		PrecMin	310	7	0	3	17	11	6	18	1	19	1	0	0	(17)
(18)		PrecStd	135	18	21	30	20	31	42	37	31	29	32	24	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	15.5	21.6	27.0	30.2	33.7	35.8	36.7	33.4	26.6	20.7	14.3	(19)
(20)		MCWB	9.4	10.2	13.1	16.7	18.3	21.3	21.6	21.0	19.7	16.9	14.0	10.6	(20)	
(21)		2%	DB	9.2	13.4	18.3	25.0	28.4	32.1	33.4	34.5	30.3	24.5	18.2	11.1	(21)
(22)		MCWB	7.3	10.0	11.5	15.7	18.3	21.7	21.2	20.5	19.5	16.1	12.9	8.9	(22)	
(23)		5%	DB	7.1	11.3	15.5	22.8	26.8	30.4	31.3	32.9	28.1	22.4	16.2	9.0	(23)
(24)		MCWB	6.3	8.7	9.7	14.4	17.7	20.9	20.5	20.3	18.2	15.2	12.0	7.4	(24)	
(25)		10%	DB	5.1	9.5	13.1	20.5	24.8	28.5	29.3	30.8	25.8	19.6	13.9	7.4	(25)
(26)		MCWB	4.5	7.6	8.5	13.1	17.0	20.1	19.8	20.4	17.9	14.1	10.9	6.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	11.8	14.0	18.3	21.3	24.9	23.7	23.9	21.6	18.4	14.6	10.7	(27)
(28)		MADB	12.0	14.5	20.1	24.2	24.5	30.3	27.9	30.0	30.2	23.8	19.0	13.6	(28)	
(29)		2%	WB	7.8	10.3	12.1	16.9	20.1	23.1	22.7	22.8	20.3	16.7	13.6	9.1	(29)
(30)		MADB	8.9	12.4	16.6	22.9	26.1	27.8	28.3	29.8	27.6	22.3	17.2	10.8	(30)	
(31)		5%	WB	6.3	9.0	10.6	15.5	18.9	22.0	22.0	21.9	19.4	15.8	12.5	7.7	(31)
(32)		MADB	7.2	11.5	14.8	21.3	24.5	27.6	28.6	29.2	25.8	21.0	15.9	8.9	(32)	
(33)		10%	WB	4.5	7.7	9.3	14.2	18.0	21.1	21.2	21.1	18.4	14.8	11.2	6.5	(33)
(34)		MADB	5.3	9.4	12.3	18.8	23.1	26.6	27.6	28.0	23.8	18.9	13.7	7.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.6	7.6	10.6	12.4	12.0	12.1	12.9	13.3	11.8	11.6	8.5	5.1	(35)	
(36)		5% DB	MADB	7.7	10.6	14.1	15.8	15.2	14.7	15.9	16.4	15.7	16.3	11.5	7.7	(36)
(37)		MCWBR	5.7	7.4	8.2	7.4	6.7	5.6	5.4	4.5	5.3	8.3	6.6	5.7	(37)	
(38)		5% WB	MADB	7.4	9.6	12.7	14.0	13.3	12.8	13.7	13.8	13.5	14.0	10.5	7.4	(38)
(39)	MCWBR	5.7	6.8	7.5	6.9	6.8	5.8	5.3	4.7	5.2	7.4	6.3	5.6	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.365	0.403	0.459	0.441	0.439	0.454	0.457	0.428	0.390	0.362	0.338	(40)	
(41)		taud	2.323	2.267	2.197	2.056	2.175	2.216	2.178	2.170	2.213	2.308	2.337	2.344	(41)	
(42)		Ebn at Noon	736	790	811	796	829	831	811	791	776	751	699	689	(42)	
(43)		Edh at Noon	78	100	124	156	143	138	142	138	121	95	77	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.10	1.95	3.21	4.63	5.61	6.12	6.03	5.45	3.81	2.54	1.45	0.85	(44)	
(45)		RadStd	0.12	0.30	0.44	0.44	0.53	0.45	0.48	0.51	0.42	0.27	0.13	0.12	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.70	N/A	N/A	+1.01	N/A	N/A	-125	-188	+113	+58

Nomenclature: See separate page